
The Hallucination Trilemma: A Topological Impossibility for Faithful, Covering, and Calibrated Language Models

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Abstract

We prove that no question-answering model whose confidence map is continuous on a connected question space can simultaneously satisfy three properties: *faithful* (threshold-confidence answers are strictly correct), *covering* (the model produces both correct and incorrect answers), and *strictly calibrated* (confidence above the threshold if and only if correct, and confidence below the threshold if and only if incorrect). Coverage is satisfied by any model that produces both correct and incorrect answers. The binding constraint is between faithfulness and strict calibration: calibration turns the two coverage witnesses into confidence values on opposite sides of the threshold; connectedness and continuity force a threshold-confidence question; calibration pins the truth-distance to zero there; faithfulness demands it be strictly negative. The contradiction is structural, not statistical, but it is a statement about this formal design target rather than a claim that current deployed LLMs satisfy strict biconditional calibration. As calibration is idealized toward the zero-slack biconditional form, the constraint tightens. We call this the **Hallucination Trilemma**. The same logical core applies without any topology when the question set contains a single boundary example, and the formal development includes multi-turn, stochastic, discrete, and approximate variants. The development is machine-verified in Lean 4 against Mathlib v4.28 with no axioms beyond the three standard Lean kernel axioms.

1 Introduction

Hallucination (a model producing a confident but wrong answer) is one of the most reliably observed problems with deployed language models (Maynez et al., 2020; Ji et al., 2023; Zhang et al., 2023). Most proposed fixes are empirical: train on more data, add RLHF, tune the temperature, apply a post-hoc calibration method. This paper takes a different angle. The problem is not just an engineering gap under the idealized conditions studied here. There is a mathematical obstruction, and it does not depend on how the model is trained.

The impossibility. Consider any model that takes a question and returns an answer together with a confidence score. It is natural to want three things from such a model:

- **Faithful:** when the model is at least half-confident, its answer should actually be correct.
- **Covering:** the model should sometimes be right and sometimes be wrong; it should not simply refuse every question or output a constant answer.
- **Calibrated:** confidence should track correctness in both directions. High confidence should mean the answer is right; low confidence should mean it is wrong.

For a continuous confidence map on a connected question space, these three properties cannot coexist. The proof requires no training assumptions, no architecture details, no statistics. It is an application of the intermediate value theorem.

Why. If the model is covering, there is a question q_+ where its answer is correct and a question q_- where its answer is wrong. Calibration converts these into statements about the confidence score: above $\frac{1}{2}$ at q_+ and below $\frac{1}{2}$ at q_- . If the question space is connected and the confidence map is continuous, the intermediate value theorem forces the confidence to hit $\frac{1}{2}$ at some question, call it q_0 . Calibration then forces the model’s answer at q_0 to sit on the truth boundary. But faithfulness says: if you are at least half-confident, you must be strictly correct. At q_0 the model is exactly half-confident, which is enough for faithfulness to fire, and the answer is on the boundary, which is not strictly correct. Contradiction.

What the result says about real models. Coverage is satisfied by any model that produces both correct and incorrect answers: every useful LLM produces both correct and incorrect answers. The binding question is calibration. The theorem does not say that current deployed LLMs satisfy all three conditions and derive a contradiction; no current model satisfies strict biconditional calibration. What it says is that the formal design target (a model that is simultaneously faithful, strictly calibrated, covering, and continuous in the confidence coordinate on a connected question space) is unreachable. As a model is idealized toward zero-slack calibration, the constraint between calibration and faithfulness tightens. Section 4.5 makes this precise: under ε -calibration and ε -coverage, a threshold-confidence question has truth-distance in $[-\varepsilon, 0)$. At zero slack this becomes the exact contradiction $0 \leq \delta_M(q_0) < 0$. The obstruction is structural, not a matter of insufficient training.

What distinguishes this from prior work. Xu et al. (2024) prove hallucination via computability: a finite model cannot represent all computable functions, so out-of-distribution inputs will always find gaps. That argument concerns model capacity at the distribution boundary and says nothing about calibration. Karpowicz (2025) prove a conflict among four conditions (truthful response generation, semantic information conservation, relevant knowledge revelation, knowledge-constrained optimality) via mechanism design and proper scoring rules. Neither proof uses the IVT; neither makes calibration a central condition.

The present result is distinguished by three things: the conditions are different (three, with calibration explicit), the proof uses only IVT and connectedness, and it is machine-verified. It is, to our knowledge, the first hallucination impossibility formalized in a proof assistant.

Contributions.

- **The Hallucination Trilemma** (Theorems 3.1–3.2): faithful, covering, and calibrated are mutually inconsistent for any continuous model on a connected question space. Each of the three conditions is individually necessary.
- **Discrete core** (Theorem 4.3): on any set with no topology at all, a single boundary question suffices to derive the same contradiction. The continuous result factors through this two-line logical core; IVT is used only to produce the boundary question, not to deliver the contradiction.
- **Extensions:** multi-turn conversations (Theorem 4.1), stochastic models with a positive-probability hallucination guarantee (Theorem 4.2), and an antipodal variant where coverage follows from the structure of the question space.
- **Approximate bridge** (Section 4.5): under ε -calibration, the impossibility becomes quantitative. A threshold-confidence question q_0 exists with $\delta_M(q_0) \in [-\varepsilon, 0)$; improving calibration tightens this margin. This connects the idealized result to approximate real models and localises the failure to the in-distribution/OOD transition band (Corollary 4.6).
- **Lean formalization:** all results are machine-verified in Lean 4 against Mathlib v4.28, no `sorry`, headline theorems dependent only on the three standard Lean kernel axioms.

2 Setup

We work with a minimal abstract model of a question-answering system. The formalism is general enough to cover any model that returns an answer alongside a confidence score, whether that score comes from a softmax probability, a calibrated estimator, or a learned scalar head.

The model. Let Q be the *question space*, a connected topological space, and A the *answer space*. A *model* is a continuous map $M: Q \rightarrow A \times \mathbb{R}$. The two components are:

$$\text{ans}(q) = (M(q))_1 \in A, \quad \text{conf}(q) = (M(q))_2 \in \mathbb{R}.$$

Connectedness of Q is the key topological assumption; it says the question space is not split into separated pieces. The Lean proof uses connectedness (more precisely, preconnectedness of the whole space) and continuity of the confidence coordinate; it does not require a path between every two questions. Path-connected embedding spaces are important examples, but the theorem is stated at the weaker connectedness level. When Q is genuinely discrete, two-sided coverage alone does not force a threshold crossing; the discrete result of Section 4.3 applies instead when a boundary question is present.

Measuring correctness. Rather than a binary right/wrong label, we use a continuous *truth-distance* $\delta: Q \times A \rightarrow \mathbb{R}$. The sign encodes correctness: $\delta(q, a) < 0$ means the answer a is correct at question q , $\delta(q, a) > 0$ means it is wrong, and $\delta(q, a) = 0$ is the boundary between the two. We write $\delta_M(q)$ for the truth-distance of the model’s own answer: $\delta_M(q) = \delta(q, \text{ans}(q))$.

This signed-distance formulation is general. It covers continuous relaxations of binary correctness, soft factuality scores, semantic similarity thresholds, and other scoring functions with a natural zero. A discontinuous two-valued label $\delta = \pm 1$ on a connected domain cannot itself satisfy both continuity and two-sided coverage; in that case the discrete theorem, or a continuous surrogate score, is the relevant formal object.

The threshold $\frac{1}{2}$. The conditions use $\frac{1}{2}$ as the boundary between high and low confidence. This is conventional: it is the natural 50/50 point where the model considers itself equally likely to be right or wrong, but not mathematically necessary. Replacing $\frac{1}{2}$ with any constant $c \in \mathbb{R}$ throughout gives an equivalent theorem with an identical proof: coverage produces witnesses above and below c , connectedness forces a crossing at c , calibration pins $\delta_M = 0$ there, faithfulness demands $\delta_M < 0$ there. The impossibility holds at whatever threshold the three conditions share.

Why these three conditions. The three conditions are the tightest set that creates the impossibility: each pair of two is simultaneously achievable (Section 3.1); all three together are not (Theorem 3.2).

Coverage is a weak condition. It requires only that the model produce at least one correct answer and at least one wrong answer across its question domain. It is the hypothesis that gives the IVT argument its two witnesses q_+ and q_- ; without it, the confidence map need not cross the threshold at all. In the proof it is load-bearing even though in practice any model that produces both correct and incorrect answers satisfies it.

The three conditions. We now define the three properties precisely.

Definition 2.1 (Faithfulness). M is faithful if, for every question q :

$$\text{conf}(q) \geq \frac{1}{2} \implies \delta_M(q) < 0.$$

When the model is at least half-confident, its answer is strictly correct.

Definition 2.2 (Coverage). M satisfies coverage if there exist questions $q_+, q_- \in Q$ with $\delta_M(q_+) < 0$ (a correct answer) and $\delta_M(q_-) > 0$ (a wrong answer). The model is not trivially correct everywhere or trivially wrong everywhere.

Definition 2.3 (Strict calibration). M is strictly calibrated if, for every question q :

$$\text{conf}(q) > \frac{1}{2} \iff \delta_M(q) < 0 \quad \text{and} \quad \text{conf}(q) < \frac{1}{2} \iff \delta_M(q) > 0.$$

Confidence above $\frac{1}{2}$ is exactly equivalent to being correct; confidence below $\frac{1}{2}$ is exactly equivalent to being wrong.

Strict calibration is a biconditional between the sign of the confidence relative to the threshold and the sign of the truth-distance. It is different from the usual statistical notion of calibration (where $\text{conf}(q)$ approximates the probability of correctness), and is best read as an idealized threshold calibration condition. It is the condition that gives the IVT argument its bite: it pins the truth-distance to zero wherever the confidence crosses $\frac{1}{2}$.

3 The Hallucination Trilemma

The argument has two steps. First we show that coverage and calibration together force a “boundary question”: a question where the model sits exactly halfway between confident and uncertain, and whose answer lands exactly on the truth boundary. Then faithfulness fires at that question and produces a contradiction.

Theorem 3.1 (Boundary question). *Let Q be connected and M be a continuous model with continuous truth-distance δ . If M is covering and strictly calibrated, then there is a question $q_0 \in Q$ where $\text{conf}(q_0) = \frac{1}{2}$ and $\delta_M(q_0) = 0$.*

Proof. Coverage gives q_+ with $\delta_M(q_+) < 0$ and q_- with $\delta_M(q_-) > 0$. Applying strict calibration at each witness: $\text{conf}(q_+) > \frac{1}{2}$ and $\text{conf}(q_-) < \frac{1}{2}$.

The confidence map $q \mapsto \text{conf}(q)$ is continuous on the connected space Q . By the intermediate value theorem, it must take the value $\frac{1}{2}$ somewhere; call that question q_0 .

At q_0 the confidence is exactly $\frac{1}{2}$, so it is neither above nor below. The two biconditionals of strict calibration then both fire in the negative direction: $\delta_M(q_0)$ is not < 0 (from the first biconditional) and not > 0 (from the second). Therefore $\delta_M(q_0) = 0$. $\square$

Theorem 3.2 (Hallucination Trilemma). *No continuous model on a connected question space is simultaneously faithful, covering, and strictly calibrated.*

Proof. By Theorem 3.1, there is a question q_0 with $\text{conf}(q_0) = \frac{1}{2}$ and $\delta_M(q_0) = 0$. Since $\text{conf}(q_0) = \frac{1}{2} \geq \frac{1}{2}$, faithfulness requires $\delta_M(q_0) < 0$. But $\delta_M(q_0) = 0$. Contradiction. $\square$

The proof uses no facts about the model’s architecture, training procedure, or parameter count. It uses only the continuity of M and the connectedness of Q . Any model (transformer, MLP, nearest-neighbor, or retrieval-augmented) that satisfies the three conditions would have to violate one of these two properties.

Tightness. The impossibility is tight: relaxing the faithfulness threshold by any $\varepsilon > 0$ breaks it. The model $Q = [0, 1]$, $\text{conf}(q) = 1 - q$, $\delta_M(q) = q - \frac{1}{2}$ satisfies strict calibration and coverage. It satisfies $(\frac{1}{2} + \varepsilon)$ -faithfulness ($\text{conf} \geq \frac{1}{2} + \varepsilon \Rightarrow q \leq \frac{1}{2} - \varepsilon \Rightarrow \delta_M \leq -\varepsilon < 0$) for every $\varepsilon > 0$, but fails standard faithfulness at $q = \frac{1}{2}$ where $\text{conf} = \frac{1}{2}$ and $\delta_M = 0$. A model satisfying $(\frac{1}{2} + \varepsilon)$ -faithfulness provides no faithfulness guarantee for questions with confidence in $[\frac{1}{2}, \frac{1}{2} + \varepsilon]$; that threshold band is the price of escaping while preserving strict calibration and coverage.

3.1 Each hypothesis is necessary

Figure 1 (left) presents the trilemma with three vertices: Faithfulness, Calibration, and Coverage, all under the assumption of a continuous model on a connected space. Each of the three is individually necessary: the three counterexamples below show that dropping any one allows the other two to be satisfied. Separately, the exact faithfulness threshold is load-bearing: §3 shows that raising the threshold by any $\varepsilon > 0$ allows strict calibration and coverage to coexist with the weakened faithfulness condition.

Covering + Calibrated + Continuous, not Faithful. Take $Q = [0, 1]$, $\text{conf}(q) = 1 - q$, $\delta_M(q) = q - \frac{1}{2}$. Strict calibration holds: $\text{conf} > \frac{1}{2} \iff q < \frac{1}{2} \iff \delta_M < 0$, and $\text{conf} < \frac{1}{2} \iff \delta_M > 0$. Coverage holds: $q = \frac{1}{4}$ gives $\delta_M < 0$; $q = \frac{3}{4}$ gives $\delta_M > 0$. Faithfulness fails at $q = \frac{1}{2}$: $\text{conf} = \frac{1}{2} \geq \frac{1}{2}$ but $\delta_M = 0$, not < 0 . (This is the same model used for the tightness argument in §3.)

Faithful + Calibrated + Continuous, not Covering. Take any model with constant $\text{conf} = \frac{3}{4}$ and $\delta_M(q) < 0$ everywhere. Faithfulness: $\frac{3}{4} \geq \frac{1}{2} \Rightarrow \delta_M < 0$. Calibration: $\text{conf} > \frac{1}{2}$ and $\delta_M < 0$ are both always true, so the first biconditional holds; the second ($\text{conf} < \frac{1}{2} \iff \delta_M > 0$) holds vacuously since both sides are always false. Coverage fails: no wrong-answer witness q_- exists.

Faithful + Covering + Calibrated, not Continuous. On $Q = [0, 1]$ with $Q^+ = (\frac{1}{2}, 1]$ and $Q^- = [0, \frac{1}{2})$, set $\text{conf}(q) = \frac{3}{4}$ on Q^+ and $\text{conf}(q) = \frac{1}{4}$ on Q^- , with $\delta_M < 0$ on Q^+ and $\delta_M > 0$ on Q^- . Faithfulness, coverage,

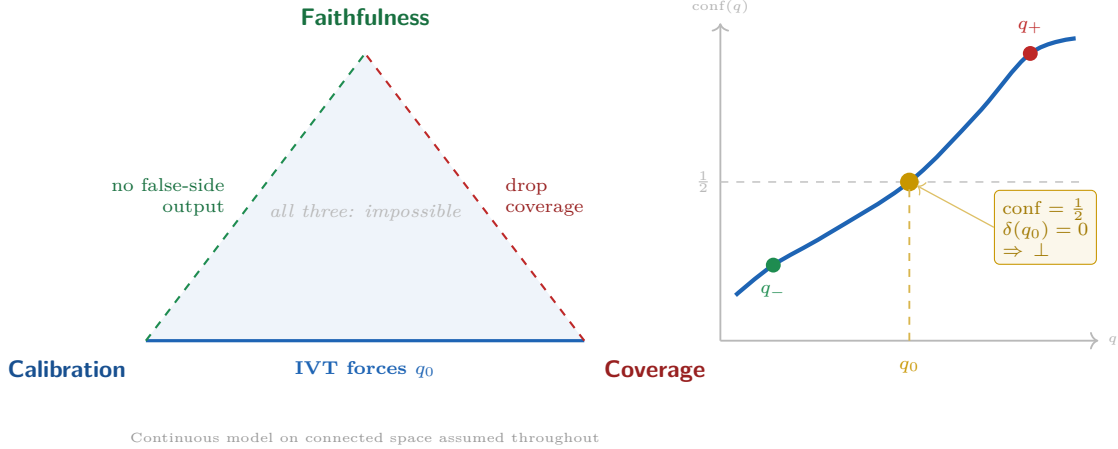


Figure 1: **Left:** The Hallucination Trilemma. For any continuous model on a connected space, no model satisfies all three. The blue (solid) edge is the main result; dashed edges correspond to counterexamples (§3.1). **Right:** The IVT argument. Coverage gives q_- (confidence $< \frac{1}{2}$) and q_+ (confidence $> \frac{1}{2}$); connectedness forces the crossing q_0 ; calibration pins $\delta(q_0) = 0$; faithfulness then demands $\delta(q_0) < 0$ – contradiction.

and strict calibration all hold pointwise. However, the confidence map has a jump discontinuity at $q = \frac{1}{2}$ and so does not satisfy the theorem’s continuity hypothesis. The model is not continuous, not the three conditions failing; continuity is what the impossibility requires.

4 Extensions

Three natural objections to the core result suggest possible escapes. This section closes all three.

Objection	Proposed escape	Result
Single-turn only	History-dependent $M : Q^T \rightarrow A \times \mathbb{R}$	Q^T connected; trilemma applies
Deterministic	Stochastic model	Dichotomy: wrong answers w.p. > 0
Connected space	Discrete, no boundary q^*	Genuine escape (all three hold)
	Discrete, with boundary q^*	Same contradiction; no topology needed

4.1 Multi-turn conversations

Objection: A model that sees prior conversation turns can use context to avoid the boundary question by identifying that q_0 is dangerous and respond differently based on history.

Why it fails: The history space Q^T is connected whenever Q is connected, since a finite product of connected spaces is connected. The trilemma applies to $M : Q^T \rightarrow A \times \mathbb{R}$ without modification. The boundary question now exists in history space rather than single-turn space, but it cannot be avoided. Context is not the issue; topology is.

Theorem 4.1 (Multi-turn trilemma). *If Q is connected, any continuous, faithful, covering, and calibrated history-dependent model $M : Q^T \rightarrow A \times \mathbb{R}$ does not exist.*

Proof. Q^T is connected. Apply Theorem 3.2. □

A secondary observation: any t -turn history space can be injectively embedded into a t' -turn history space (for $t' > t$) by padding with a fixed question (HoF_11.adversary_boundary_reachable). This is a structural

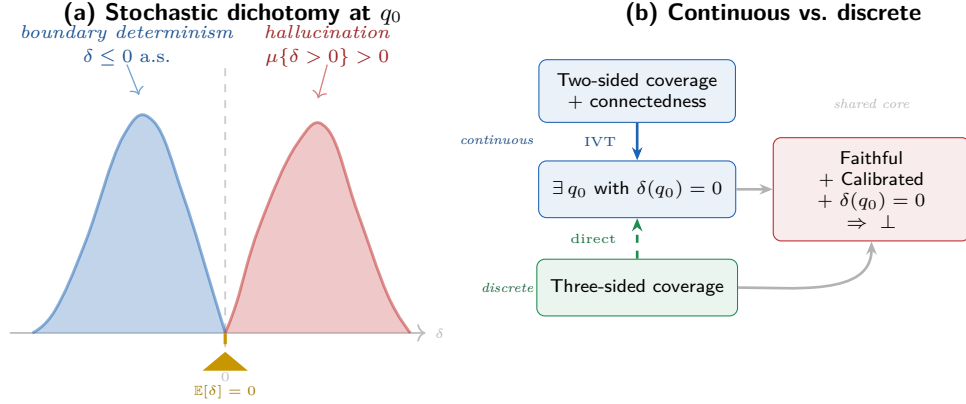


Figure 2: **(a)** At the forced boundary point q_0 where $\mathbb{E}[\delta] = 0$, a confident stochastic model faces a dichotomy: all mass at $\delta = 0$ (boundary determinism, blue) or positive mass at $\delta > 0$ (hallucination, red). The balance constraint forces one or the other. **(b)** The continuous trilemma and the discrete trilemma share a two-line logical core (right). They differ only in how the boundary witness q_0 is produced: IVT from two-sided coverage (blue) vs. explicit three-sided coverage (green).

embedding of history spaces; whether a particular model preserves the trilemma hypotheses after padding is a model-specific question.

4.2 Probabilistic models

Objection: A randomized model can spread probability mass across multiple answers at the boundary question, avoiding any single committed wrong output.

Why it fails: Randomization does not remove the boundary question. The expected-value trilemma applies to the expected maps and yields a q_0 with $\mathbb{E}[\delta_M(q_0)] = 0$. The stochastic dichotomy (Theorem 4.2) then shows that any model answering with positive probability at or above the confidence threshold at q_0 must produce wrong answers there with positive probability.

A stochastic model draws an answer-confidence pair (a, c) from a distribution $M(q)$ rather than returning a single value. There are two ways to extend the trilemma to this setting.

The first is immediate. Let $\bar{c}(q) = \mathbb{E}_{M(q)}[c]$ and $\bar{\delta}(q) = \mathbb{E}_{M(q)}[\delta(q, \cdot)]$ be the expected confidence and expected truth-distance. Assume both maps are continuous from Q to $\mathbb{R}$ (this requires the distribution $M(q)$ to vary continuously with q in an appropriate sense, e.g., weakly). If $(\bar{c}, \bar{\delta})$ satisfy the three conditions in expectation, the trilemma applies directly via Theorem 3.2, yielding q_0 with $\bar{c}(q_0) = \frac{1}{2}$ and $\bar{\delta}(q_0) = 0$.

The second result is sharper. It weakens faithfulness from expectations to individual samples, and shows that randomization at q_0 cannot help.

Theorem 4.2 (Stochastic dichotomy). *Let (Ω, μ) be a probability space and $c, d: \Omega \rightarrow \mathbb{R}$ measurable functions with d integrable. Suppose $\int d \, d\mu = 0$, individual-sample faithfulness holds μ -almost everywhere ($c(\omega) \geq \frac{1}{2} \Rightarrow d(\omega) < 0$), and $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$.*

Proof. Suppose $\mu\{d > 0\} = 0$, so $d \leq 0$ μ -a.e. Then $-d \geq 0$ a.e. and $\int (-d) \, d\mu = 0$. A non-negative integrable function with zero integral equals zero almost everywhere (`integral_eq_zero_iff_of_nonneg_ae`), so $d = 0$ μ -a.e. But on the positive-probability set $\{c \geq \frac{1}{2}\}$, sample faithfulness gives $d < 0$ a.e. A positive-probability set cannot simultaneously satisfy $d < 0$ a.e. and $d = 0$ a.e. Contradiction. $\square$

A stochastic model at q_0 with positive-probability high-confidence outputs must produce wrong answers at q_0 with positive probability. Figure 2(a) shows the dichotomy geometrically.

4.3 Discrete question sets

Objection: Real models operate on finite token vocabularies. Token space is discrete, not connected, so the IVT does not apply and the topological argument is irrelevant to deployed systems.

Answer: The discrete case splits into two sub-cases. If the model has no question with $\delta_M(q) = 0$ (no boundary question), all three conditions can hold simultaneously: this is a genuine escape. If the model has at least one boundary question, the IVT is not needed: the same two-line contradiction fires directly, with no topology at all.

The key observation is purely logical. Suppose a question q^* exists where the model’s answer lands exactly on the truth boundary: $\delta_M(q^*) = 0$. Strict calibration immediately squeezes the confidence to $\frac{1}{2}$: the first biconditional says confidence is above $\frac{1}{2}$ only if $\delta < 0$, which fails; the second says confidence is below $\frac{1}{2}$ only if $\delta > 0$, which also fails. So $\text{conf}(q^*) = \frac{1}{2}$. But faithfulness says confidence $\geq \frac{1}{2}$ requires $\delta < 0$. Contradiction.

Theorem 4.3 (Discrete impossibility). *Let Q and A be arbitrary sets with no topology. No model M can simultaneously be faithful, strictly calibrated, and have a question $q^* \in Q$ with $\delta_M(q^*) = 0$.*

This requires no finiteness, no paths, no topology. It is two lines of arithmetic. The continuous trilemma factors through it: the IVT supplies the boundary question from two-sided coverage and connectedness; the discrete result then delivers the contradiction. On a truly disconnected set, coverage alone cannot force a boundary question into existence, so the discrete version requires one as an explicit assumption – “three-sided coverage” ($\exists q^+, q^0, q^-$ with $\delta < 0, \delta = 0, \delta > 0$ respectively). Figure 2(b) shows the two paths to the same logical core.

A clean characterisation follows: under strict calibration, a model is faithful if and only if it has no boundary questions at all. Faithfulness is exactly the condition that excludes $\delta = 0$.

4.4 Antipodal variant

When questions come in natural “opposite” pairs (negated premises, reversed polarity, switched yes/no framing), the question space may admit a continuous involution $\sigma: Q \rightarrow Q$ with $\sigma(\sigma(q)) = q$ and $\sigma(q) \neq q$. If the truth-distance reverses sign under this flip ($\delta_M(\sigma(q)) = -\delta_M(q)$), then the intermediate value theorem forces a zero without any coverage hypothesis at all: just pick any q , observe that $\delta_M(q)$ and $\delta_M(\sigma(q))$ have opposite signs, and invoke connectedness. Coverage is then automatic from the antipodal structure rather than an assumption.

4.5 Approximate calibration: connecting the theorem to real models

The exact theorem requires strict biconditional calibration. No real model satisfies this. This section proves a quantitative version that applies to models with a one-way approximate calibration condition, and shows how the exact contradiction is recovered at zero slack.

Two-sided ε -calibration. For $\varepsilon \geq 0$, define a model as ε -calibrated at threshold c if:

$$\delta_M(q) < -\varepsilon \Rightarrow \text{conf}(q) > c + \varepsilon \quad \text{and} \quad \delta_M(q) > \varepsilon \Rightarrow \text{conf}(q) < c - \varepsilon.$$

This is the one-way part of threshold calibration needed by the proof: very-correct answers force confidence clearly above c , and very-wrong answers force confidence clearly below c . Inside the truth-distance band $|\delta_M(q)| \leq \varepsilon$, no constraint is imposed by this definition.

Theorem 4.4 (Approximate Hallucination Trilemma). *Let $\varepsilon \geq 0$. Let Q be connected and let the confidence map of M be continuous. If M satisfies ε -coverage ($\exists q_+$ with $\delta_M(q_+) < -\varepsilon$ and $\exists q_-$ with $\delta_M(q_-) > \varepsilon$), two-sided ε -calibration, and c -faithfulness ($\text{conf}(q) \geq c \Rightarrow \delta_M(q) < 0$), then there exists q_0 with $\text{conf}(q_0) = c$ and $\delta_M(q_0) \in [-\varepsilon, 0]$.*

Proof. ε -calibration converts the coverage witnesses: $\delta_M(q_+) < -\varepsilon$ gives $\text{conf}(q_+) > c + \varepsilon \geq c$, hence $\text{conf}(q_+) > c$, and $\delta_M(q_-) > \varepsilon$ gives $\text{conf}(q_-) < c - \varepsilon \leq c$, hence $\text{conf}(q_-) < c$. IVT gives q_0 with $\text{conf}(q_0) = c$.

Lower bound $\delta_M(q_0) \geq -\varepsilon$: if $\delta_M(q_0) < -\varepsilon$, ε -calibration gives $\text{conf}(q_0) > c + \varepsilon$, contradicting $\text{conf}(q_0) = c$.
Upper bound $\delta_M(q_0) < 0$: from c -faithfulness since $\text{conf}(q_0) = c \geq c$. $\square$

Lean: `HoF_12.approx_trilemma`.

At $\varepsilon = 0$, the conclusion becomes $0 \leq \delta_M(q_0) < 0$, an immediate contradiction. The zero-slack contradiction is stated formally as `HoF_12.exact_from_approx`. Strict calibration and ordinary coverage imply the zero-slack approximate hypotheses at threshold $\frac{1}{2}$; the formal bridge theorem is `HoF_12.exact_from_strictCalibrated_via_approx`.

Discrete band constraint. The same arithmetic applies to any discrete domain without IVT:

Theorem 4.5 (Discrete band constraint). *Let $\varepsilon \geq 0$ and let Q, A be arbitrary sets. If M satisfies two-sided ε -calibration and c -faithfulness, then for every question q^* with $c \leq \text{conf}(q^*) \leq c + \varepsilon$:*

$$\delta_M(q^*) \in [-\varepsilon, 0).$$

Proof. $\delta_M(q^*) < 0$ follows from faithfulness. $\delta_M(q^*) \geq -\varepsilon$: if $\delta_M(q^*) < -\varepsilon$, ε -calibration gives $\text{conf}(q^*) > c + \varepsilon$, contradicting $\text{conf}(q^*) \leq c + \varepsilon$. $\square$

Lean: `HoF_12.discrete_approx_bridge` and `HoF_12.no_boundary_in_upper_band`.

Interpretation. The two theorems give the precise connection between the idealized result and real models:

- On a *connected continuous* domain, ε -coverage forces a threshold-confidence question to exist (IVT). That question has $\delta_M \in [-\varepsilon, 0)$: the model is correct there, but only barely.
- On a *discrete* domain, the band question may or may not exist. If it does, the same upper-band constraint holds. If it does not, this particular approximate argument does not apply; the exact discrete impossibility still applies whenever a boundary question exists.
- As calibration improves ($\varepsilon \rightarrow 0$), the allowed margin at the threshold-confidence question shrinks to zero. At zero slack the formal conclusion is contradictory.

Improving calibration and improving faithfulness are not independent: any gain in calibration (smaller ε) tightens the faithfulness constraint at the threshold-confidence question by that amount.

Application to out-of-distribution inputs. Let Q_{in} be the in-distribution question set and Q_{ood} the OOD set, both subsets of a connected ambient question space Q . Suppose the model is ε -correct in-distribution ($\delta_M(q) < -\varepsilon$ for $q \in Q_{\text{in}}$) and ε -wrong OOD ($\delta_M(q) > \varepsilon$ for $q \in Q_{\text{ood}}$). Under two-sided ε -calibration, this gives $\text{conf} > c + \varepsilon$ on Q_{in} and $\text{conf} < c - \varepsilon$ on Q_{ood} .

Corollary 4.6 (OOD band failure). *These are exactly the ε -coverage witnesses for Theorem 4.4. The resulting q_0 satisfies $\text{conf}(q_0) = c$ and $\delta_M(q_0) \in [-\varepsilon, 0)$. Since $\text{conf} > c + \varepsilon$ on Q_{in} and $\text{conf} < c - \varepsilon$ on Q_{ood} , q_0 belongs to neither: it lies in the transition region of the ambient space, outside the two sets on which the model is assumed clearly correct or clearly wrong.*

The failure is not at an arbitrary question but at the transition region between in-distribution and OOD behavior. As $\varepsilon \rightarrow 0$, the allowed margin shrinks to zero; at zero slack the exact contradiction fires at the forced threshold point.

5 Lean Formalization

All results in this paper are machine-verified in Lean 4 against Mathlib v4.28. The complete Lean source is included in the arXiv submission zip (`HallucinationProofs/`, `CCHProofs/`, `Foundation/`, and `ManifoldProofs/` directories). The artifact consists of thirteen files under the `HallucinationProofs` namespace. Each file imports only `Mathlib` and can be checked independently. Running `lake build` against Mathlib v4.28 completes with zero errors and no `sorry` placeholders. The `HoF_FinalVerification` file

runs `#print axioms` on all headline theorems; every result reduces to at most `{propext, Quot.sound, Classical.choice}` – the standard Lean kernel axioms.

The files follow the logical order of the paper. `HoF_01–06` build the vocabulary (spaces, truth-distance, the three definitions). `HoF_07` contains the core impossibility, using Mathlib’s `IsPreconnected.intermediate_value2` for the IVT step and two `linarith` calls for the calibration squeeze. `HoF_08` handles the antipodal variant. `HoF_09` contains path-based discrete results (`discrete_sign_change`, `discrete_hallucination`) that precede the pure-set formulation; `HoF_10` contains the no-topology discrete impossibility of Section 4.3. `HoF_11` handles multi-turn and stochastic extensions (using `Pi.connectedSpace` and `integral_eq_zero_iff_of_nonneg_ae`), and `HoF_12` the approximate bridge (Section 4.5), including `exact_from_strictCalibrated_via_approx`, which derives the exact strict-calibration contradiction from the zero-slack approximate theorem.

An annotated proof sketch is in Section B.

6 Discussion

6.1 The three available relaxations

The trilemma identifies three conditions that cannot simultaneously hold. Each can be relaxed; the theorem says one of them must be. The three relaxations correspond to the three edges of the trilemma triangle (Figure 1).

Drop coverage. A model that abstains on questions where it is uncertain (returning “I don’t know” rather than an answer) violates Coverage. Selective prediction (Kamath et al., 2020; Feng et al., 2020) formalizes this. It is achievable; the trilemma says Coverage is what must be traded to make the other two conditions hold simultaneously.

Relax faithfulness. Accept that the guarantee starts only above the boundary threshold. The trilemma guarantees the exact obstruction: at least one question q_0 exists where confidence equals c and the answer lands on the truth boundary. A model that relaxes faithfulness by threshold ε (requiring only confidence $\geq c + \varepsilon$ to guarantee correctness) can coexist with strict calibration and coverage, but provides no guarantee for questions with confidence in $[c, c + \varepsilon)$.

Weaken calibration. Strict biconditional calibration can be replaced by the one-way approximate calibration condition of Section 4.5. Then the exact contradiction becomes a margin statement: $\delta_M(q_0) \in [-\varepsilon, 0)$ at a threshold-confidence question. At zero slack this reduces to $0 \leq \delta_M(q_0) < 0$, recovering the exact contradiction.

6.2 Optimal relaxation under deployment constraints

The three relaxations are not equivalent in cost, and the choice between them is not arbitrary—it is determined by the deployment context and the asymmetry of error types.

The cost asymmetry. Dropping Coverage through abstention eliminates wrong answers at the cost of utility. A model that refuses to answer whenever it is uncertain satisfies Faithfulness and Calibration trivially, but provides no value on the hard questions where it is most needed. The practical cost is proportional to how often the deployed model encounters questions near the confidence boundary in its actual operating environment. Relaxing Faithfulness accepts that some confident answers will be wrong; in practice, the frequency of wrong confident answers depends on how densely the operational query distribution falls near that boundary. This relaxation imposes a *localized* cost: the model remains reliable away from the boundary and unreliable only in the uncertain middle zone. Weakening Calibration changes the informativeness of the confidence signal used across the question space—a potentially *global* cost rather than a local one. A model with weaker calibration gives downstream systems less formal justification for interpreting confidence scores as truth indicators, which matters when those scores are used for routing, human review escalation, or risk thresholding.

The relaxations are not symmetric. Abstention is a decision made at inference time on a question-by-question basis and can be applied selectively. Faithfulness relaxation is a tolerance on output quality near the threshold. Calibration weakening affects how confidence scores should be interpreted wherever they are consumed. A practitioner choosing among the three is not choosing between equivalent trade-offs with different labels: they are choosing between an inference-time coverage cost, a threshold-band faithfulness cost, and a confidence-signal interpretability cost.

Boundary density is the key empirical unknown. All three costs depend on the same underlying quantity: how frequently the deployed model encounters questions near the confidence boundary in actual operation. The trilemma guarantees at least one such question exists on a connected question space, but says nothing about how common such questions are in practice. If the operational distribution concentrates on questions where the model is clearly competent or clearly incompetent—and the two populations are well separated—all three relaxation costs are low and the practical severity of the trilemma is limited. If the operational distribution regularly places queries in the uncertain middle zone, the costs are high and the choice of relaxation becomes critical. This connects the impossibility theorem to a concrete empirical research agenda: measuring how often a deployed model operates near its confidence boundary—on representative operational data, not benchmark data—is the quantity that determines which relaxation is cheapest for a given system.

Domain context determines the dominant relaxation. In high-stakes domains where a wrong confident answer is catastrophic (medical diagnosis, legal reasoning, safety-critical control), abstention is the correct relaxation: the cost of refusing to answer is recoverable; the cost of a confident wrong answer may not be. In high-throughput domains where refusing to answer significantly degrades utility (search, code completion, customer service), faithfulness relaxation may be preferred if boundary questions are rare in the operational distribution. In domains where confidence scores are consumed by downstream systems for routing or risk assessment (scientific literature review, financial analysis, automated triage), calibration weakening is the most dangerous relaxation, because it silently corrupts the signal that downstream components depend on.

Connection to the out-of-distribution result. Theorem 4.6 identifies where boundary questions tend to live: at the transition between in-distribution and out-of-distribution behavior. A model that is clearly correct on its training distribution and clearly wrong on far-out-of-distribution inputs will concentrate its boundary questions in the transition zone between the two. How wide that zone is, and how often operational queries fall into it, is an empirical property of the deployment. Systems deployed in stable, well-bounded operational environments will have narrow transition zones and low boundary density. Systems regularly asked questions that push against the edges of their training will have wide transition zones and high boundary density, and will feel the trilemma more acutely.

6.3 Comparison with prior impossibilities

Table 1: Comparison with prior hallucination impossibility results.

	Xu et al. (2024)	Karpowicz (2025)	This work
Proof technique	Computability	Mech. design + scoring rules	Topology (IVT)
Architecture-independent	✓	✓	✓
Applies to OOD inputs	✓	×	×
Applies to in-dist. structure	×	✓	✓
Addresses calibration explicitly	×	×	✓
Number of conditions	N/A	4	3
Machine-verified	×	×	✓

The Xu et al. obstruction is about model capacity; this one is structural. Xu et al. (2024) show hallucination is inevitable because a finite model cannot represent all computable functions, so out-of-distribution inputs will always expose gaps. The argument concerns a model’s capacity relative to the function class it is asked to approximate; it applies at the distribution boundary. The present result concerns

the interaction between the structure of the question space, continuity of the confidence map, and the calibration and coverage hypotheses. The boundary question q_0 is forced independently of training procedure or parameter count.

The Karpowicz obstruction uses four conditions; this one uses three. Karpowicz (2025) prove that four properties conflict – truthful response generation, semantic information conservation, relevant knowledge revelation, and knowledge-constrained optimality, for any model performing knowledge aggregation. The proof uses mechanism design (Green-Laffont), proper scoring rules (Savage), and Jensen’s inequality on transformer log-sum-exp aggregation as three independent proof paths; the first two are not architecture-specific. The present result derives a contradiction from three conditions via IVT alone. The conditions are different: calibration is central here but not among Karpowicz’s four.

Calibration is the central condition. Neither prior result makes calibration the key variable. Here it is: the biconditional form of calibration is exactly what pins the truth-distance to zero at the forced crossing point. A model that does not satisfy strict biconditional calibration escapes the exact impossibility; the approximate bridge in Section 4.5 states the corresponding zero-slack limit.

6.4 Limitations

Continuity and connectedness. Token space is discrete; real confidence maps over raw token strings are not continuous in the topological sense. The continuous theorem should therefore be read as applying to a connected semantic or embedding-level abstraction with a continuous confidence coordinate. More directly, the discrete result (Theorem 4.3) requires no topology: one boundary question suffices for the same contradiction. Continuity is used to derive the existence of q_0 from two-sided coverage; it is not needed once a boundary question is assumed.

Existence vs. reachability. The theorem says q_0 exists; it does not say how easy it is to find. The boundary question is the point where $\text{conf}(q_0) = c$ and $\delta_M(q_0) = 0$; how frequently such a question appears in practice, and how hard it is to reach adversarially, are empirical questions this paper does not address. Coverage is the hypothesis that allows IVT to produce q_0 from two witnesses; it is weak (any model with at least one correct and one wrong answer satisfies it) but load-bearing in the proof.

Strict calibration and real models. No deployed model satisfies the strict biconditional exactly. The theorem is a statement about the formal design space: the region satisfying all three conditions is empty under the stated continuity and connectedness assumptions. The approximate trilemma (Section 4.5) extends this to models with ε -calibration: the failure margin at the threshold-confidence question is at most ε , tightening as calibration improves.

No empirical validation. The paper does not exhibit q_0 for any deployed model, measure the width of the uncertainty band in practice, or compare how far current models are from each condition. These are open empirical questions.

Broader Impact Statement

The result establishes an existence proof. The boundary question q_0 is shown to exist under the three stated conditions; it is not constructed and does not constitute a capability uplift. The theorem applies to models satisfying strict biconditional calibration; claims about real deployed systems require a separate argument connecting approximate calibration to the idealized conditions here.

7 Related Work

Empirical hallucination. Maynez et al. (2020) identify faithfulness failures in summarization; Ji et al. (2023) survey the problem across tasks; Zhang et al. (2023) give a taxonomy. These papers measure and categorize hallucination but do not ask whether it is structurally unavoidable. The present result gives a structural answer: it is, for any model satisfying the three natural desiderata.

Theoretical impossibilities. Xu et al. (2024) prove hallucination is inevitable via computability: finite models cannot represent all computable functions, so out-of-distribution inputs will always expose gaps. The result concerns model capacity relative to the function class; it says nothing about calibration.

Karpowicz (2025) prove that four properties conflict – truthful response generation, semantic information conservation, relevant knowledge revelation, and knowledge-constrained optimality, for any model performing knowledge aggregation. The proof uses mechanism design and proper scoring rules (both general) and Jensen’s inequality on log-sum-exp aggregation (as a specific instantiation). The four conditions differ from the three here; calibration is not among them. The present result uses only IVT and three conditions, one of which is explicitly calibration.

Calibration. Guo et al. (2017) show modern networks are miscalibrated; post-hoc methods (Platt, 1999; Zadrozny & Elkan, 2001) improve this empirically. The structural interpretation of that miscalibration, namely that strict biconditional calibration is incompatible with faithfulness and coverage, is new to this paper. The calibration literature treats the problem as correctable; the trilemma says correction must come at the cost of one of the other two properties.

Selective prediction. Kamath et al. (2020); Feng et al. (2020) study abstention as a way to avoid high-confidence errors. The trilemma gives this a formal grounding: abstention is exactly dropping Coverage, and it is the one unambiguously conservative choice.

Defense Trilemma. Anonymous (2025) apply the same IVT machinery to prompt-injection defenses. That paper shows no continuous defense can be simultaneously utility-preserving, complete, and continuous. The present paper shows the analogous result for truthfulness: the same topological engine, applied to the confidence map rather than the defense map, with faithfulness playing the role of utility preservation and calibration providing the forcing condition.

Formalization. Avigad (2024) and Buzzard et al. (2020) demonstrate that substantial mathematical results can be machine-verified in Lean. This paper is, to our knowledge, the first hallucination impossibility result to be formalized in a proof assistant.

8 Conclusion

No model with a continuous confidence map on a connected question space is simultaneously faithful, covering, and strictly calibrated. The proof uses only the intermediate value theorem and the definition of the three conditions. Conversation context does not remove the obstruction when the history space remains connected, and randomization yields a positive-mass wrong-answer conclusion under the sample-level hypotheses of Theorem 4.2. Discrete token space is a genuine escape from the IVT step unless a boundary question is present; once such a question exists, the same two-line logical contradiction applies without topology.

The tightness result (§3) shows that raising the faithfulness threshold by any $\varepsilon > 0$ lets strict calibration and coverage coexist with the weakened guarantee. The approximate trilemma (Section 4.5) gives the corresponding zero-slack picture for ε -calibration: at $\varepsilon = 0$ the formal conclusion is $0 \leq \delta_M(q_0) < 0$.

All results are in Lean 4, sorry-free, against Mathlib v4.28.

Ethics Statement

This work is a theoretical study of impossibility results for language model calibration and hallucination. It introduces no new datasets, models, or systems. The results are mathematical theorems with machine-verified Lean proofs. We identify no direct ethical risks from the work itself. The findings may inform how practitioners select among trade-offs when deploying language models in high-stakes domains, which we regard as a positive contribution to responsible deployment.

Reproducibility Statement

All results stated in this paper are formally verified in Lean 4 against Mathlib v4.28. The Lean source files are included in the supplementary material and will be released publicly upon acceptance. The main theorem (Theorem 3.2), the tightness result (§3), and the discrete analogue (Theorem 4.3) are all sorry-free. Proof sketches for the remaining corollaries are provided in Section 5. No empirical experiments are reported; there is no stochastic component to reproduce.

References

- Anonymous. The defense trilemma: A topological impossibility for prompt-injection defenses. *Under review*, 2025.
- Jeremy Avigad. Mathematics and the formal turn. *Bulletin of the American Mathematical Society*, 61(2): 225–240, 2024.
- Kevin Buzzard, Johan Commelin, and Patrick Massot. Formalising perfectoid spaces. *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 299–312, 2020.
- Shi Feng, Zhiyuan Feng, Bing Xu, Deli Zhao, Yang Liu, and Sheng Li. Language generation with recurrent neural networks on Wikipedia. *arXiv preprint arXiv:2011.01085*, 2020.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. *International Conference on Machine Learning*, pp. 1321–1330, 2017.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.
- Amita Kamath, Robin Jia, and Percy Liang. Selective question answering under domain shift. *arXiv preprint arXiv:2006.09462*, 2020.
- Piotr Karpowicz. On the fundamental impossibility of hallucination control in large language models. *arXiv preprint arXiv:2506.06382*, 2025.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. *arXiv preprint arXiv:2005.00661*, 2020.
- John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pp. 61–74, 1999.
- Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. Hallucination is inevitable: An innate limitation of large language models. *arXiv preprint arXiv:2401.11817*, 2024.
- Bianca Zadrozny and Charles Elkan. Obtaining calibrated probability estimates from decision trees and naive Bayesian classifiers. In *International Conference on Machine Learning*, pp. 609–616, 2001.
- Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. Siren’s song in the ai ocean: A survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219*, 2023.

A Full Proof of Theorem 3.2

We reproduce the full proof of Theorem 3.2 with all sub-steps made explicit.

Proof of Theorem 3.2. Suppose for contradiction that M is faithful, covering, and strictly calibrated.

Step 1: Extract confidence witnesses. By Coverage, there exist $q_+, q_- \in Q$ with $\delta_M(q_+) < 0$ and $\delta_M(q_-) > 0$. By Strict Calibration (first biconditional) at q_+ : $\delta_M(q_+) < 0 \Rightarrow \text{conf}(q_+) > \frac{1}{2}$. By Strict Calibration (second biconditional) at q_- : $\delta_M(q_-) > 0 \Rightarrow \text{conf}(q_-) < \frac{1}{2}$.

Step 2: Apply IVT. The map $q \mapsto \text{conf}(q) = (M(q))_2$ is continuous as a component of the continuous map $M: Q \rightarrow A \times \mathbb{R}$. The space Q is connected. We have $\text{conf}(q_+) > \frac{1}{2} > \text{conf}(q_-)$. By the intermediate value theorem, there exists $q_0 \in Q$ with $\text{conf}(q_0) = \frac{1}{2}$.

Step 3: Pin truth-distance at the crossing. We show $\delta_M(q_0) = 0$. If $\delta_M(q_0) < 0$, then by Strict Calibration (first biconditional) $\text{conf}(q_0) > \frac{1}{2}$, contradicting $\text{conf}(q_0) = \frac{1}{2}$. If $\delta_M(q_0) > 0$, then by Strict Calibration (second biconditional) $\text{conf}(q_0) < \frac{1}{2}$, again a contradiction. Hence $\delta_M(q_0) = 0$.

Step 4: Contradiction with Faithfulness. Since $\text{conf}(q_0) = \frac{1}{2} \geq \frac{1}{2}$, Faithfulness demands $\delta_M(q_0) < 0$. But Step 3 gives $\delta_M(q_0) = 0$. Contradiction. $\square$

B Lean Proof Sketch

The following is the core Lean 4 proof of `hallucination_trilemma` from `HoF_07_TrilemmaCore.lean`, lightly annotated.

```
theorem hallucination_trilemma
  {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
  [TopologicalSpace A]
  (M : Q → A × ℝ) (delta : Q × A → ℝ)
  (hM_conf : Continuous (fun q => (M q).2))
  (hF : TrilemmaFaithful M delta)
  (hC : TrilemmaCovering M delta)
  (hCal : StrictCalibrated M delta) : False := by
-- Step 1: extract q0 with conf = 1/2 and delta = 0
obtain <q0, hconf, hd> :=
  hallucination_trilemma_strict M delta ... hC hCal
-- Step 2: Faithfulness at q0 gives delta < 0
have hge : (M q0).2 >= 1/2 := ge_of_eq hconf
have hlt : delta (q0, (M q0).1) < 0 := hF q0 hge
-- Step 3: contradiction
linarith
```

The helper `hallucination_trilemma_strict` applies `IsPreconnected.intermediate_value2` to the confidence map between the q_+ and q_- witnesses from Coverage, obtaining q_0 with $\text{conf}(q_0) = \frac{1}{2}$, then uses the biconditionals of Strict Calibration to force $\delta(q_0) = 0$.